

How Green Factories Improve Firm ESG Performance?

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Abstract

This paper explores the role of Green Factory Certification (GFC) as a driver of firms' environmental, social, and governance (ESG) performance in building a green manufacturing system. Using GFC as a quasi-natural experiment, the study employs a staggered difference-in-differences approach to examine its causal effect on ESG outcomes among A-share listed manufacturing firms in China from 2009 to 2021. Results show that GFC significantly enhances ESG performance, with the strongest effects on the environmental dimension, followed by social and governance aspects. Findings remain robust under synthetic control estimation and heterogeneity tests. Mechanism analysis identifies four channels through which GFC improves ESG performance: alleviating financing constraints, promoting green technology innovation, increasing media attention, and mitigating managerial myopia. Heterogeneity analysis shows that larger firms and those with higher environmental awareness experience a more pronounced positive effects. These results highlight the vital role of GFC in advancing firm ESG performance.

Keywords: Green manufacturing; Green Factory Certification; Firm ESG Performance; Staggered Difference-in-Differences (DID) Method

1 Introduction

With the advancement of industrialization, global challenges like climate change, air pollution, water scarcity, and soil degradation have reached critical levels, threatening human survival and sustainable development. To address these issues, a green transformation in economic growth model that balances economic development and environmental protection has become crucial, aligning with the shared goals of the international community. Firms, as key economic participants, have the responsibility to driving this green transition. Since the UN introduced the concept of Environmental, Social, and Governance (ESG) framework in 2004, an increasing number of firms have recognized the role of ESG in sustainable development and adopted ESG oriented management strategies. Current research has shown that strong ESG performance can promote green innovation, improve financial performance, reduce enterprise risks, and enhance enterprise value (Wang et al., 2023a; Chen et al., 2023; Bissoondoyal-Bheenick et al., 2023; Albuquerque et al., 2019). In this context, advancing firm ESG performance through effective policy intervention has become an urgent and far-reaching task.

As a modern manufacturing model characterized by low consumption, low emissions, high efficiency, and high benefits, the establishment of green manufacturing systems has become a focal point in the global pursuit of green development. It is also a landmark policy initiative for governments around the world to actively promote sustainable development. The *National Advanced Manufacturing Strategy* (NAMS) proposed by the United States in 2022 and the *Corporate Sustainability Reporting Directive* (CSRD) implemented by the European Union in 2023 both explicitly require companies to undergo green transformation. As a major emerging market and developing country, China places significant emphasis on green manufacturing system development. In May 2015, the State Council issued *Made in China*

2025, the action plan for the first decade of China’s strategy to become a leading manufacturing power, which explicitly advocates for an efficient, clean, low-carbon, and circular green manufacturing system. In September 2016, the Ministry of Industry and Information Technology (MIIT) released the *Notice on Promoting Green Manufacturing System Development*, proposing to build a green manufacturing framework centered around green products, green factories, green industrial parks, and green supply chain management enterprises. Among these, green factories serve as the primary entities of green manufacturing and form the core support units within the green manufacturing system. Green factories, particularly those in traditional sectors like steel and chemicals, have become priority targets for green factory recognition due to their high pollution, high energy consumption, significant industry influence, high public attention, and strong policy pressure, and have received targeted support to lead the broader green transformation of other industries. In theory, the establishment of green factories influences firm ESG performance by leveraging incentive and constraint mechanisms that generate positive externalities for both the economy and the environment. However, the question of whether green factories certification can enhance firm ESG performance and the mechanisms through which this occurs remains unresolved.

To address this issue, this paper uses Green Factory Certification (GFC) as a quasi-natural experiment, analyzing data from A-share listed manufacturing firms in China (2009–2021). A staggered difference-in-differences (DID) model assesses the impact of GFC on ESG performance, mechanisms, and firm-level heterogeneity. The paper’s contributions are summarized in three key aspects:

First, this paper uncovers the “black box” of how green manufacturing promotes ESG performance. While previous literature has examined the effects of green manufacturing on ESG performance, the specific policies driving these impacts remain unclear. Existing research on policies influencing corporate ESG performance often focuses on individual measures, such as green credit policies (Lei et al., 2023b; Gao and Liu, 2023), government subsidies (He et al., 2023), and environmental regulation (He et al., 2022). However, there is a notable gap in studies exploring the combined effects of policy packages on ESG performance. The green factory certification policy, which offers not only financial rewards but also incentives such as preferential production and credit policies, represents a typical policy package within the green industry. With its authoritative and detailed certification standards, this paper specifically examines the impact of green factory certification and quantifies its effect on corporate ESG performance, providing valuable insights for policy development.

Second, this article expands the research on green factories. While some studies have examined the effects of green factory certification on stock prices, technological innovation, and information disclosure (Chen et al., 2022; Zhu et al., 2023; Wang et al., 2024), they have not directly addressed the specific impact of certification on corporate ESG performance and its individual dimensions. This article investigates the mechanisms through which green factory certification influences ESG performance, focusing on both internal and external factors. It demonstrates how certification can alleviate financing constraints, foster green technology innovation, boost media attention, correct managerial myopia, and ultimately enhance corporate ESG performance. This study not only fills the gap in understanding the mechanisms underlying the impact of green factory certification on ESG performance but also offers fresh perspectives and recommendations for optimizing green factory certification policies.

Third, in terms of research methodology, this paper employs an Staggered Difference-in-Differences (DID) approach. On the one hand, the green factory certification process is standardized by government policies, meaning that firms do not have full control over their certification status, ensuring randomness. On the other hand, the certification policy is implemented gradually across different regions and years, the staggered DID method effectively addresses the heterogeneous impacts of policy rollout at various time points, overcoming the potential biases of traditional DID approaches in handling phased policies. This methodology allows for more accurate estimation of the impact of green factory certification on corporate ESG performance, thereby strengthening the reliability and academic contribution of the findings.

2 Literature Review

The current literature provides extensive insights into green factories and their diverse impacts. As noted by [Wei et al. \(2024\)](#), 'green factory' certification functions as a discretionary environmental mechanism designed to advance long-term environmental policies. Therefore, in order to achieve sustainable development, green factory certification plays an important role in balancing economic benefits and environmental protection. In terms of economic effects, [Li et al. \(2024\)](#) observe that Green Factory Certification (GFC) enhances commercial credit availability by reducing information asymmetry. [Ye et al. \(2022\)](#) leverages the Green Manufacturing Engineering Implementation Guide, demonstrating that green industrial policies boost total factor productivity (TFP) in manufacturing firms significantly. [Wei et al. \(2024\)](#) observes that GFC significantly improve labor productivity. Additionally, [Chen et al. \(2022\)](#) highlights the positive impact of GFC on stock prices.

From an environmental perspective, [Wang et al. \(2024\)](#) finds that green factories improve environmental information disclosure quality through pollution and emission reduction efforts, along with increased engagement from green investors and institutional investors' on-site evaluations. The research of [Zhang et al. \(2022\)](#) shows that green factories have a positive impact on corporate green innovation behavior. Although the above literature shows that green factories significantly influence the economic and environmental benefits of firms, research on their impact on ESG performance remains limited.

The factors that affect the ESG performance of enterprises can be roughly divided into internal factors and external factors. On the internal factors side, the impact of firm characteristics and managerial attributes on ESG performance. Research has shown that factors such as CEO gender ([Aabo and Giorici, 2023](#)), CEO green experience ([Huang and Wei, 2023](#)), ownership structure ([Wang et al., 2023b](#)), equity incentives ([Zeng et al., 2023](#)), managerial myopia ([Liu and Zhang, 2023](#)), firm size ([Drempetic et al., 2020](#)), digital transformation ([Fang et al., 2023](#)), green bond issuance ([Zheng et al., 2023](#)), and technological innovation ([Huang et al., 2023](#)) significantly impact ESG performance.

For external factors, some scholars examine the influence of capital markets or government actions on ESG performance. Studies highlight the role of institutional investors ([Wang et al., 2023c](#)), stock market liberalization ([Wang et al., 2023d; Yin et al., 2023](#)), bond markets ([Jiang et al., 2023](#)), media coverage ([He et al., 2024](#)), local government debt pressures ([Nie et al., 2023](#)), and government environmental focus ([Liu et al., 2024](#)). In addition, the research also includes the impact of environmental policies on ESG outcomes. Findings indicate that China's Environmental Protection Tax Law ([He et al., 2023](#)), the Central Environmental Inspection Policy ([Wang et al., 2023a](#)), and green credit policies ([Nie et al., 2023; Gao and Liu, 2023](#)) all influence firm ESG performance.

The existing literature provides valuable insights into the economic and environmental impacts of Green Factory Certification (GFC), but research on its effects on overall ESG (Environmental, Social, and Governance) performance remains limited. Most existing research focuses on the individual economic and environmental benefits of GFC, but there is a need to integrate these effects into a unified framework that captures GFC's role in improving ESG performance as a whole. Moreover, while studies have explored various internal and external factors influencing firm ESG performance, such as managerial attributes, ownership structure, and government policies, there is limited exploration of how a policy intervention like GFC can specifically enhance ESG outcomes across the board. This gap in the literature underscores the need for further investigation into the mechanisms through which GFC influences not only the environmental dimension but also the social and governance aspects of ESG performance. Understanding these mechanisms will help policymakers and firm leaders better leverage green certifications to promote sustainable development and achieve broader ESG objectives. Thus, this paper aims to fill this gap by exploring how GFC improves firm ESG performance.

3 Policy background and hypothesis

3.1 Policy background

In May 2015, the State Council launched *Made in China 2025* initiative, introducing the concept of green factories with goals centered on intensified manufacturing, the use of non-toxic raw materials, clean production processes, efficient resource utilization of waste, and low-carbon energy consumption. In September 2016, MIIT, along with the NDRC, MST, and MOF, issued the Green Manufacturing Engineering Implementation Guide (2016–2020) to pilot a green manufacturing system, targeting the creation of 1,000 demonstration factories by 2020. From 2017 to 2023, MIIT identified 5,113 national-level green factories across eight batches, spanning traditional industries (cement, steel, textiles, chemicals) and new sectors (energy-saving equipment, new energy production).

In terms of regulatory mechanisms, the green factory certification adopts a dynamic list management system, conducting random inspections of certified green factories. Enterprises that fail to meet the required standards during these inspections are removed from the demonstration list. In February 2023, based on this dynamic management requirement, the Ministry of Industry and Information Technology (MIIT) removed 41 companies from the green factory list. Furthermore, MIIT mandates that certified green factories regularly self-declare their green manufacturing performance indicators on the public green manufacturing service platform, showcasing advanced green manufacturing practices and typical experiences. Additionally, companies are encouraged to disclose green information and publish annual reports on their green and low-carbon development.

Regarding incentive and support mechanisms, certified green factories are eligible for various government incentives at multiple levels. At the central level, MIIT provides financial support to green factories through industrial transformation and upgrading funds and special construction funds. At the local level, governments offer a range of policy benefits to certified green factories, including one-time cash subsidies, green credit, tax reductions, government procurement preferences, priority in administrative approvals, exemption from staggered production requirements, and power usage exemptions. For instance, Hangzhou provides subsidies of 2 million RMB and 1 million RMB for companies awarded national and provincial-level green factory certification, respectively. Zhenjiang offers a special credit line of 10 billion RMB for green factory construction, with interest rate discounts of 15% and 10% for national and provincial green factory awardees, respectively. Suzhou Industrial Park grants exemptions for orderly power usage to certified green factories.

These comprehensive policies and incentives have laid a strong foundation for accelerating the green transformation of China’s manufacturing sector and promoting the widespread adoption of sustainable production practices.

3.2 Research Hypothesis

3.2.1 The effect of GFC on firm ESG performance

Modern firm theory posits that companies should not only generate profits for shareholders but also consider the interests of stakeholders, in line with ESG principles. However, in practice, the externalities and substantial financial investments associated with ESG activities can increase production costs and reduce short-term profitability, leading to a diminished willingness among companies to proactively engage in ESG practices. As a significant green industrial policy, GFC encourages companies to transition toward environmentally friendly and low-carbon practices, thereby enhancing firm ESG performance through the use of policy incentives and regulatory frameworks. Specifically, the impact of GFC on firm ESG performance is evident across its three key dimensions.

From an environmental performance perspective, GFC enhances firms' environmental awareness by requiring them to meet rigorous standards set by the central government. By July 2023, the MIIT had established 82 specific industrial standards for green factories, ensuring strict compliance that heightens environmental consciousness among firm management. Certified factories benefit from government support, including tax breaks, subsidies, and rewards, which incentivize green investments and technological advancements. This support motivates firms to upgrade their equipment, enhance resource efficiency, and reduce pollutant emissions, ultimately improving firm environmental performance.

From a social performance perspective, GFC significantly enhances firm social responsibility. The certification process requires companies to assume social responsibilities in areas such as production safety, product quality, employee training, employee welfare, and community engagement, thereby improving overall firm social performance. First, ensuring production safety and enhancing product quality are key components of GFC. Second, to meet the certification requirements, firms must regularly provide environmental awareness and skill training to employees, thereby boosting their environmental knowledge and capabilities. Additionally, firms are encouraged to improve employee welfare, which in turn increases job satisfaction and fosters a sense of belonging among employees. Lastly, GFC promotes community interaction and cooperation, enhancing firm social influence and brand value through activities such as charitable donations, environmental advocacy, and volunteer work. These initiatives enable companies to more effectively fulfill their social responsibilities.

From a governance performance perspective, GFC has the potential to enhance firm management systems. Unlike traditional firm governance, ESG focused governance emphasizes the integration of environmental and social considerations into firm governance frameworks, mechanisms, and decision-making processes, thereby improving overall management systems. GFC prioritizes management systems as essential benchmarks, encouraging firms to adopt practices like quality management, environmental stewardship, occupational health and safety protocols, energy efficiency measures, and more. This not only standardizes business operations and strengthens internal management but also mitigates environmental and social risks, thereby improving firm governance capabilities. Consequently, GFC contributes to the enhancement of firm governance performance. Based on these insights, the hypothesis is:

H1: GFC enhances firm ESG performance.

3.2.2 Potential mechanisms

Alleviate firm financing constraints. Firm behavior is significantly shaped by financing capabilities (Nie et al., 2023). Compared to conventional investment projects, ESG-related investments often involve longer cycles, substantial capital requirements, and limited returns (Taghizadeh-Hesary and Yoshino, 2019), which reduces incentives for companies to engage in ESG investments and hinders green and low-carbon development. GFC provides financial support through policy incentives such as fiscal subsidies, credit preferences, and tax reductions, easing firm financing constraints and thus improving ESG performance. Companies certified as green factories initially receive special subsidies from the MIIT and one-time financial rewards from various levels of government, increasing their internal funding. This alleviates financial burdens, reduces the cost of ESG investments, and bolsters enthusiasm for environmental protection initiatives.

Moreover, bank credit remains a vital source of external financing for companies in China (Wang et al., 2021). However, significant information asymmetry between banks and firms limits firm external financing capabilities, thereby affecting their ESG practices. Local governments encourage financial institutions to offer credit preferences and green credit support to companies with GFCs, providing indirect government support. This reduces information asymmetry between banks and firms, enabling companies to secure more favorable loan terms. Lastly, GFC helps companies build a positive green reputation, attracting

the attention and approval of external investors by reducing information asymmetry. This improves the stability of external funding and further alleviates firm financing constraints.

Improve levels of green innovation within firms. Green technological innovation is a key driver of ESG performance, as it improves resource utilization efficiency and reduces pollution emissions, enabling companies to achieve green transformation more effectively and adhere to the principles of sustainable development (Lei et al., 2023a). At the policy level, the MIIT has established firm standards that exceed existing requirements for companies seeking GFC. As of July 2023, a total of 82 industry-specific evaluation standards for green factories have been issued. To meet the requirements for GFC, companies voluntarily integrate environmental costs into their business decisions, increase investment in green technology, and elevate their green development standards.

Once certified, companies remain under the supervision of regulatory authorities. Should they fail to comply with environmental standards or experience significant environmental incidents, the government will revoke their certification in accordance with dynamic management protocols. This motivates companies to actively pursue green initiatives to secure and maintain long-term competitive advantages in the green market. Furthermore, to uphold their green reputation, companies are motivated to continuously pursue green technological innovation, reinforcing their leadership in environmental sustainability and remaining competitive with other environmentally conscious firms. Additionally, financial incentives provided by various levels of government, along with green credit support from financial institutions, help alleviate the financing constraints associated with green innovation, further incentivizing companies to pursue advancements in green technology.

Increase media attention. GFC, as a national-level honor, serves as recognition of industrial firms that lead in energy conservation, environmental protection, and green development. Thus, receiving GFC sends a strong positive signal about firms' commitment towards environmental issues. After a firm receives GFC, related media often report on the awardee, highlighting its achievements and practices in environmental protection and sustainable development, thereby raising its public and market visibility and reputation. Meanwhile, media play a crucial role in information production, dissemination, and public opinion guidance, exerting an external governance effect that can influence firms' operational and governance decisions (Yu et al., 2011). Through transparent communication on social media, firms can better convey their accomplishments in environmental protection, philanthropy, and poverty alleviation to communities and the government. This transparency enhances trust with stakeholders, boosts firm social reputation, and encourages firms to satisfy their environmental, social, and governance responsibilities, ultimately improving ESG performance (Fang et al., 2023).

Overcome managerial myopia. Managers often display shortsighted decision-making, prioritizing immediate profits and shareholder returns over long-term growth and sustainability (Stein, 1989). Myopic managers tend to favor projects with shorter durations and higher immediate returns (Holmström, 1999), while long-term investments with lower short-term returns, such as ESG initiatives, are often overlooked. And GFC can help improve such situations.

First, stewardship theory suggests that managers, driven by a need for achievement, gain intrinsic satisfaction through social recognition. Green factory certification (GFC) can encourage managers to seek moral fulfillment and social approval (Francoeur et al., 2017; Sun et al., 2024), which helps correct their myopic behavior. By positioning environmental and sustainable development as core assessment criteria, GFC increases management's focus on these issues. To gain broader recognition and demonstrate value, managers are more likely to incorporate these criteria into strategic planning and decision-making, thereby mitigating myopic behavior.

Second, certified green factories are required to disclose environmental information, increasing transparency and reducing opportunities for profit inflation and other forms of myopic actions. This enhanced transparency encourages management to adopt sustained, proactive measures to improve ESG perfor-

mance, as it builds trust within the organization and enhances their personal and firm image. As a result, managers become more committed to the firm’s long-term sustainable development.

Third, firms that receive GFC benefit from government incentives, which help alleviate short-term production pressures and increase the returns on environmental and sustainability investments. These incentives also support management in shifting focus from short-term profits to long-term sustainability, further correcting myopic behavior. The research framework is shown as Fig 1:

H2a: GFC enhances firm’s ESG performance by alleviating its financing constraints.

H2b: GFC enhances firm’s ESG performance by promoting its green innovation capabilities.

H2c: GFC enhances firm’s ESG performance by increasing media attention.

H2d: GFC enhances firm’s ESG performance by correcting managerial myopia.

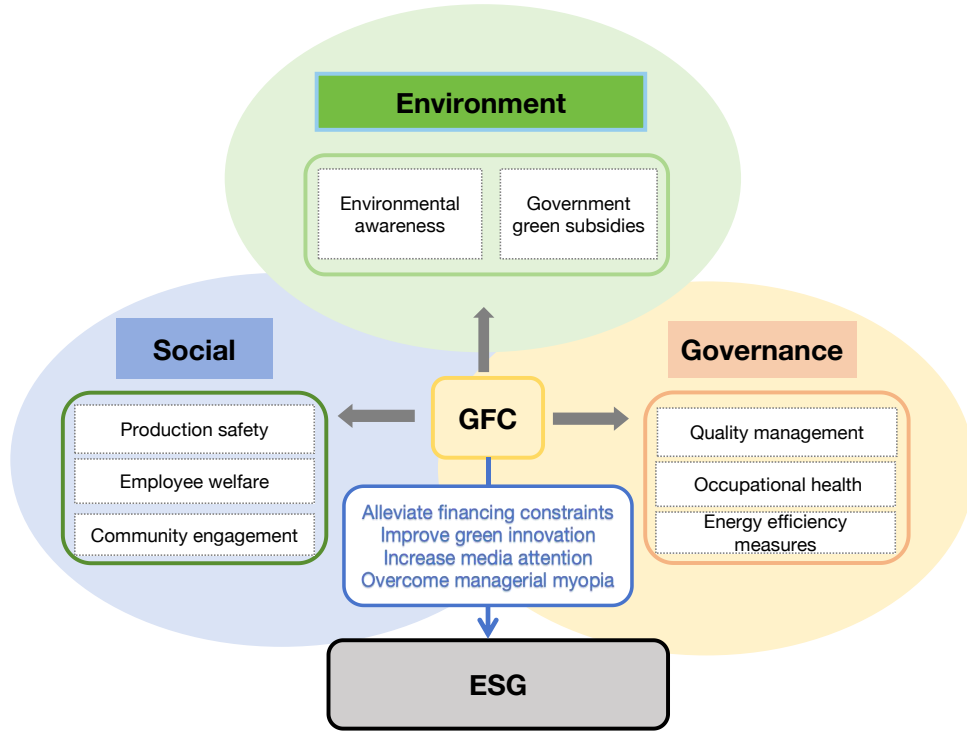


Figure 1: The research framework on the impact of GFC on firm ESG performance

4 Research design

4.1 Model setting and variables

This paper treat the GFC program, implemented since 2017, as a quasi-natural experiment. Following the methodology of Wang et al. (2024), we construct the following staggered Difference-in-Differences (DiD) model:

$$ESG_{it} = \alpha_0 + \beta GFC_{it} + \gamma X_{it} + FE + \varepsilon_{it} \quad (1)$$

In Eq.1, i denotes the listed company, t denotes the year, and ESG_{it} indicates the ESG score. We use the *Huazheng* ESG ratings data to measure the ESG performance¹. We assign a value of 1-9 to each level of the index, with higher values indicating better ESG performance. GFC_{it} indicates whether the i listed firm is recognized as a green factory in year t . We assign a value to each listed firm based on the national green factory list published by the MIIT until 2021. A listed company is assigned a value of 1 or

¹The *Huazheng* ESG Rating Index is divided into nine levels: AAA, AA, A, BBB, BB, B, CCC, CC, and C.

0 depending on whether it was certified as a green factory in the current year. The estimated coefficient β measures the effect of GFC on ESG performance.

X_{it} represents a set of company-level control variables. Drawing on relevant literature on ESG performance, we include company size, leverage ratio, book-to-market ratio, return on total assets (ROA), Tobin's Q, and the share-held by the top ten shareholders as control variables. FE represents a series of fixed effects at the firm, year, province and industry levels, controlling for unobservable factors that do not vary with firms, time, region and industry cycles. ε_{it} is the random disturbance term.

For the mechanism variables, financing constraints are represented by two indices, SA index² and FC index³. The smaller these two indices, the lower the financing constraints. The level of corporate green innovation is measured by the number of green patents, which is log-transformed after adding 1 to eliminate negative values. Media attention is assessed by following the approach of Fang et al. (2023), using the number of positive news items and the proportion of positive news provided by the China Listed Company Financial News Database (CFND). Managerial myopia is measured using the proportion of short-termism-related terms in the management discussion and analysis (MD&A) section relative to the total word count (Hu et al., 2021).

4.2 Data sources and descriptive statistics

Due to the constraints of ESG data, this study focuses on listed manufacturing firms from the A-share markets of the Shanghai and Shenzhen stock exchanges in China, covering the period from 2009 to 2021. Companies with ST and *ST⁴ listing status, firms with significant missing data, and outliers that do not comply with accounting standards are excluded from the analysis. This results in a final sample of 21,480 observations from 2,355 listed companies. ESG data is sourced from the Huazheng ESG ratings database, while the list of green factories is obtained from the official website of the MIIT and cross-matched with the listed company data by the authors. Green patent data is sourced from CNDRS⁵, and other financial data is drawn from the CSMAR⁶ database. The descriptive statistics is shown in Table 1.

Table 1: Descriptive statistics results

Variables	Observations	Mean	Std.Dev	Minimum	Maximum
ESG	21,486	4.055	1.091	1.000	8.000
E_ratings	21,486	1.943	1.161	1.000	8.000
S_ratings	21,486	4.021	1.125	1.000	9.000
G_ratings	21,486	5.359	1.405	1.000	9.000
Green Factory Certification (GFC)	21,486	0.037	0.188	0.000	1.000
FC index	19,298	0.526	0.270	0.000	0.987
SA index	21,485	3.773	0.254	2.225	5.646
Ln(Green Patent) (Lngp)	21,486	0.276	0.683	0.000	6.458
Positive News Project (Posinews)	21,413	3.168	0.905	0.000	8.287
Positive News Project proportion (Posiratio)	21,451	1.354	1.386	0.000	124.229
Myopia-related word proportion (Myopiaratio)	21,378	0.073	0.055	0.000	1.732
Company Size (Size)	21,485	21.956	1.215	16.161	27.547
Leverage Ratio (LR)	21,485	0.422	1.315	-0.195	178.346
Book-to-market Ratio (BM)	21,054	0.582	0.236	0.005	1.463
Return on Total Assets (ROA)	21,485	0.045	0.824	-30.688	108.366
TobinQ	21,054	2.278	2.958	0.684	192.705
Share held by the Top Ten Shareholders (TOP10)	21,478	0.431	0.195	0.101	0.985

²SA index calculation method: $SA = -0.737 \times Size + 0.043 \times size^2 - 0.040 \times age$. Where size is the natural logarithm of the total asset size of the firm. Age is the operating year of the firm.

³The FC index refers to the index established by Fee et al. (2009) and Chen and Zheng (2020)

⁴ST (Special Treatment) refers to a special treatment of the stock transactions of listed companies with abnormal financial conditions or other irregularities, serving as a warning of significant risk. *ST (Special Situations Treatment) indicates that the company has incurred losses for three consecutive years, warning of the risk of compulsory delisting.

⁵Chinese Research Data Services Platform

⁶China Stock Market & Accounting Research Database

5 Empirical results

5.1 Baseline result

Table 2 presents the baseline result. In column (1), the estimated coefficient of GFC is significantly positive at the 1% level after control for firm, year, industry and province. After including additional control variables, the estimated coefficient of GFC is 0.366 and remains significant at the 1% level (column 2), indicating that GFC positively enhances firm ESG performance. **H1** is verified. Specifically, holding other factors constant, a listed company's ESG score increases by 0.349 upon being certified as a green factory. Following Li et al. (2016), we assess the economic significance of this result using the sample mean. The descriptive statistics indicate that the average ESG score of the sample is 4.055, meaning that GFC leads to an 8.60% increase in ESG ratings relative to the sample mean. This improvement can be attributed to the advantages associated with GFC, such as enhanced reputation, reduced financing constraints through financial rewards and credit support, and ultimately, the promotion of ESG performance.

Table 2: Baseline results

	(1)	(2)
GFC	0.366*** (0.061)	0.349*** (0.061)
Size		0.174*** (0.025)
LR		-0.015* (0.009)
BM		0.149** (0.062)
ROA		0.007 (0.008)
TobinQ		0.003 (0.003)
TOP10		-0.005 (0.088)
Constant	4.042*** (0.002)	0.146 (0.547)
Firm FE	✓	✓
Year FE	✓	✓
Industry FE	✓	✓
Province FE	✓	✓
Observations	21,480	21,039
R ²	0.538	0.540

Note: ***, **, and * indicate 1%, 5%, and 10% significance, respectively.
Robust standard errors in parentheses are clustered at the firm level.

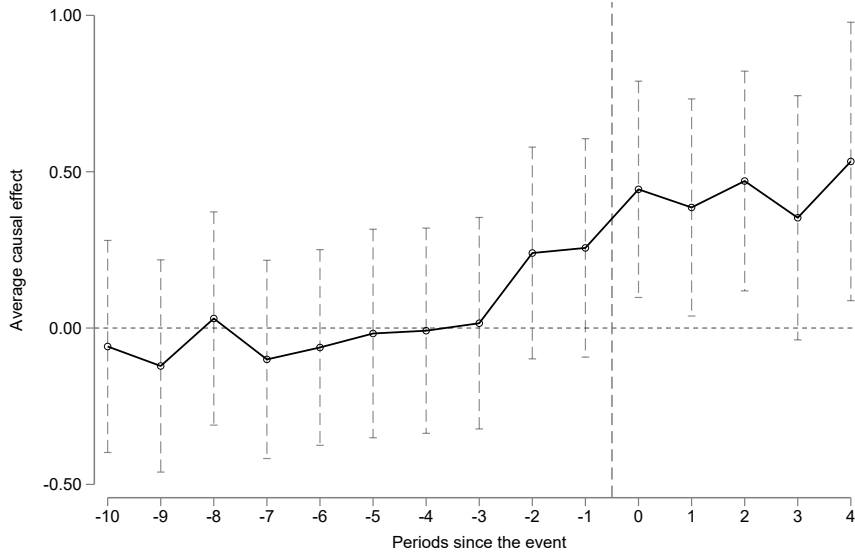
5.2 Common trend test and dynamic effects

Ensuring the validity of the common trend assumption is one of the necessary conditions for the effectiveness of the DID model, which means that green and non-green factories have the same trend in ESG performance before the GFC. Drawing on Beck et al. (2010), we examine the common trend assumption by constructing a Dynamic DID model. The specific model is as follows:

$$ESG_{it} = \alpha_0 + \sum_{k=-10}^{k=4} \beta_k GFC_{i,t_0+k} + \delta X_{it} + \gamma X_{it} + FE + \varepsilon_{it} \quad (2)$$

In Eq. (2), GFC_{i,t_0+k} is a series of dummy variables representing the k -th year when the listed company i is granted the green factory status. Specifically, t_0 represents the year when the listed company

i is awarded the green factory status, and k represents the k -th year of being granted the green factory status. The data for our paper covers the period from 2009 to 2021, including 10 years before the implementation and 4 years after the initiation of the GFC. The coefficient of interest in the equation, β_k , measures the difference in ESG ratings between the treatment group and the control group in the k -th year after green factory certification. When $k < 0$, if β_k shows no significant trend, it confirms the common trend assumption in the sample. Additionally, when $k \geq 0$, β_k captures the dynamic effects of GFC on ESG performance.



Note: The hollow circles represent the coefficients on the dashed lines denote the 95% confidence interval

Figure 2: Common trend test and dynamic effects

The common trend test (figure 2) shows that there is no significant difference in ESG performance between the treatment and control group before GFC (when $k < 0$ the coefficients are not significant), thus confirming the common trend assumption in the sample. However, there is a sustainable positive effect of the GFC on the ESG performance of firms (when $k \geq 0$ the coefficients are positive and statistically significant).

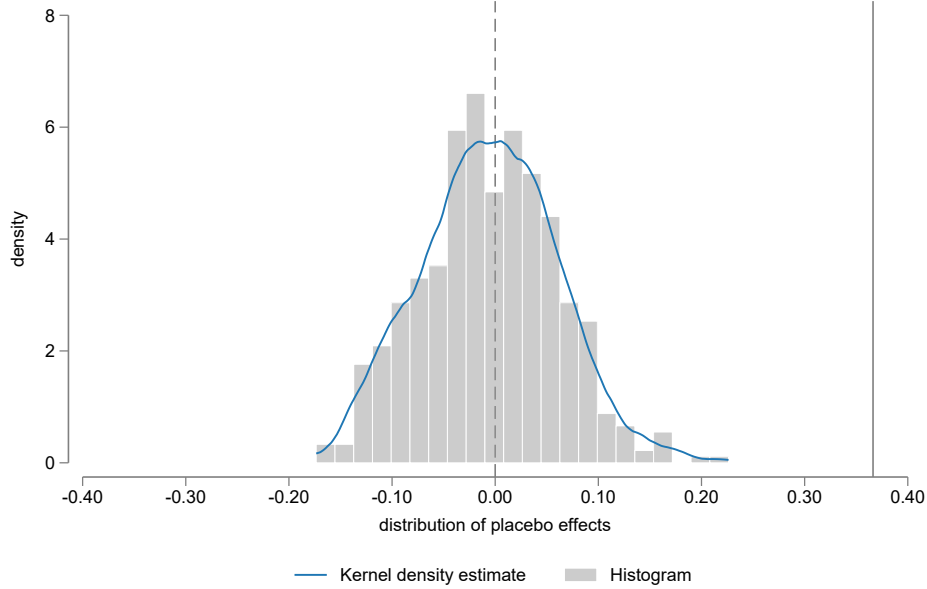
5.3 Robustness checks

We employ robustness checks by employing a placebo test, substituting regression models, accounting for confounding effects of other policies, controlling for macro factors, removing interfering samples, altering clustering methods, and using alternative estimation techniques.

5.3.1 Placebo test

Omitting variables at the firm, year, industry, and regional levels may also lead to biased estimation results. We perform a placebo test by randomly choosing the green factory by following the approach of Cai et al. (2016). The estimation includes 2533 samples, among which 252 are listed companies awarded the title of green factory. Based on this, we first randomly selects 252 listed companies from the 2533 listed companies, designating them as the "pseudo-treatment group," while other companies are designated as the control group, thus constructing pseudo green factory dummy variable GFC_i^{false} . In theory, GFC_i^{false} should not have any effect on the outcome variable, i.e., $\beta^{false} = 0$. Conversely, if β^{false} is significantly non-zero, it demonstrates the presence of identification bias in the model specification,

indicating that other unobservable factors may affect the estimation results. To avoid the influence of low-probability events, we repeat the above process 500 times, thereby generating 500 β^{false} values. The kernel density plot and histogram of the estimated coefficients for the 500 randomly generated treatment groups are shown in Figure 3). The results indicate that β^{false} is primarily centered around 0 and follows a normal distribution. Meanwhile, the benchmark estimation results are positioned at the far right of the placebo test distribution. The results of the above tests indicating that the benchmark results have not been severely affected by omitted variables.



Note: The figure shows the cumulative distribution density of the estimated coefficients if from 500 simulations randomly assigning the GFC status to the company. The rightmost vertical line indicates the result in column (2) of Table 2.

Figure 3: Placebo test.

5.3.2 Alternative specification

Considering that the dependent variable, the ESG index of listed firms, is a count variable, we also re-estimates using Poisson pseudo-maximum likelihood regressions (PPML) with multi-way fixed effects, as suggested by Cohn et al. (2022). The results shown in column (1) of Table 3 indicate that although the magnitude of the estimated coefficient of GFC has changed, it remains significant at the 1% level.

5.3.3 Eliminating the influence of other policies

One concern about accurately identifying the impact of green factory assessment on the ESG performance of listed companies is that the outcome variables are influenced by other policies during the sample period. Therefore, we further considered the impact of the implementation of green credit policies and environmental protection tax laws on the ESG performance of listed companies. Drawing on the approaches of Gao and Liu (2023), this paper constructs dummy variables for green credit policies and environmental protection tax laws, and incorporates these policy dummy variables into the benchmark model to control for their impact on the estimation results. The estimation results are shown in column (2) of Table 3. It can be observed that after controlling for policy interference, the estimated coefficient of GFC has not changed significantly, indicating that other policies during the sample period did not interfere with the benchmark results of this study.

Table 3: Robustness checks

	ESG	ESG	ESG	ESG	ESG	ESG	ESG
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
GFC	0.079*** (0.013)	0.349*** (0.061)	0.332*** (0.060)	0.306*** (0.056)	0.419*** (0.073)	0.349*** (0.056)	0.282*** (0.102)
Control variables	✓	✓	✓	✓	✓	✓	✗
Other policies	✗	✓	✗	✗	✗	✗	✗
Year×Province FE	✗	✗	✓	✗	✗	✗	✗
Year×Industry FE	✗	✗	✓	✗	✗	✗	✗
Firm FE	✓	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓	✓
Observations	21,039	21,039	21,020	21,039	16,633	21,039	8,645
R ²		0.540	0.574	0.544	0.547	0.540	

Note: ***, **, and * indicate 1%, 5%, and 10% significance, respectively. For column (1), R-squared is Pseudo R-squared. For column (6), robust standard errors in parentheses are clustered at the provincial and industry levels. For column (7), robust standard errors in parentheses are calculated using bootstrap methods, while for the remaining columns, the robust standard errors in parentheses are clustered at the firm level.

5.3.4 Controlling for macroeconomic factors

Macroeconomic factors can also cause fluctuations in ESG performance. Although the baseline model controls for industry and provincial fixed effects to account for these factors, it may not fully capture all relevant macroeconomic influences. Therefore, we incorporate industry-year and province-year interactive fixed effects into the baseline model, following the approach of [Si and Cao \(2022\)](#). The results in column (3) of Table 3 show that the estimated coefficient of GFC remains significantly positive at the 1% level, with minimal change in its value. This further reinforces the robustness of the baseline findings.

5.3.5 Outliers

We apply a 5% truncation to the dependent variable, ESG performance to mitigate the influence of outliers on the baseline results. The results, presented in column (4) of Table 3, indicate that even after implementing this procedure, the estimated coefficient for GFC remains significantly positive. Additionally, considering the disruptive effects of the epidemic in 2020 on the production, operations, and ESG practices of listed firms ([Broadstock et al., 2021](#)), we further exclude the samples from 2020 and 2021. The results in column (5) of Table 3, confirm that after these adjustments, the estimated coefficient for GFC remains significantly positive.

5.3.6 Different clustering

Considering that GFC within the same province and industry may be influenced by provincial and industry-specific environmental regulatory policies, leading to potential intra-group autocorrelation, we adopt province- and industry-double-clustered robust standard errors for testing. The results, presented in column (6) of Table 3, show that even after applying this more comprehensive clustering approach, the estimated coefficient for GFC remains positive.

5.3.7 Synthetic Control Difference-in-Differences methods

The assessment of a listed company as a green factory may be influenced by systematic differences between the treatment and control groups, particularly for companies that were not assessed as green factories. [Arkhangelsky et al. \(2021\)](#) propose the synthetic DID method by combining the staggered DID

approach with the synthetic control method (SCM). This innovative technique introduces individual and time weights, enabling a more precise alignment of pre-treatment trends and reducing the reliance on the common trend assumption inherent in traditional staggered DID models. As presented in column (7) of Table 3, the synthetic DID analysis estimates an average treatment effect of GFC on ESG performance at 0.282, which is marginally lower than the baseline value of 0.349.

5.3.8 DiD heterogeneous treatment effects

In the previous section, the staggered DID model was estimated using the two-way fixed effects model. However, some scholars have emphasized that the effectiveness of the two-way fixed effects estimator in staggered DID models is predicated on the assumption of homogeneous treatment effects. When treatment effects are heterogeneous, this kind of estimator may introduce bias, rendering the results unreliable (Huang et al., 2023).

To investigate whether the baseline findings are influenced by heterogeneous treatment effects, we follow the approach of De Chaisemartin and d'Haultfoeuille (2020) and use the *"twowayfweights"* Stata command to diagnose the weights of the two-way fixed effects model. The diagnostic results show that only 26 negative weights are present, accounting for just 3.29% of the total weights, suggesting that the bias from negative weights is minimal. We apply the stacked regression estimator proposed by Cengiz et al. (2019) to validate the robustness of the findings. Figure 4 presents the results from the stacked regression estimators, which align with the common trend test. There is almost no treatment effect before policy implementation, while ESG performance significantly improves in the pilot year when companies are certified as green factories. These findings suggest that the baseline results are unaffected by heterogeneous treatment effects.

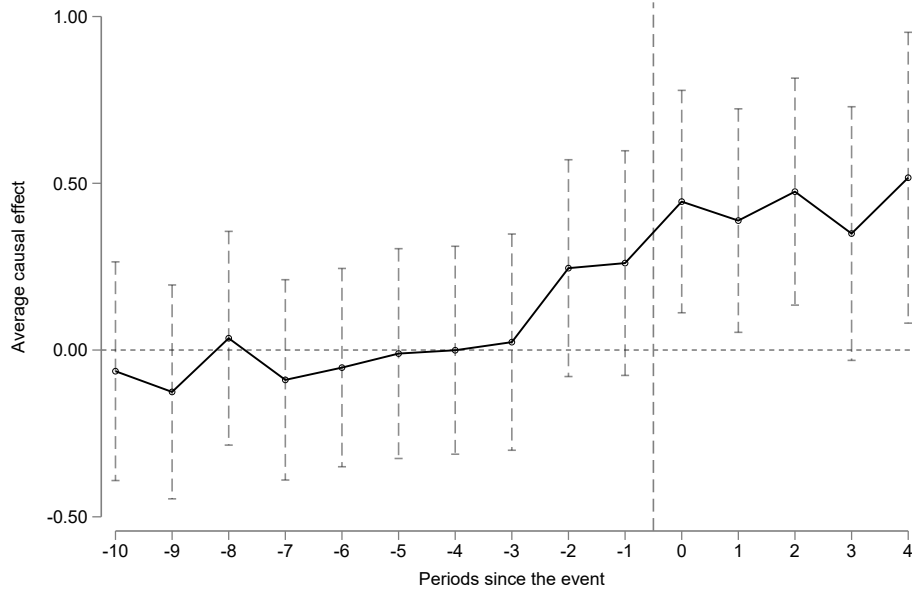


Figure 4: Event study

5.4 Mechanism analysis

In the baseline regression, we found that GFC significantly enhances firms' ESG performance. In mechanism analysis section, we first analyze the influence of GFC on individual ESG sub-indicators and then explore the potential channels through which GFC influences firms' ESG scores.

5.4.1 Impact of GFC on ESG sub-indicators

We begin by examining how GFC influences the performance of company ESG sub-indicators, thereby impacting overall ESG performance. Referring to the construction method of *Huazheng* ESG Index, this paper constructs three sub items: Environmental (E_ratings), Social (S_ratings), and Governance (G_ratings). Table 4 shows that the estimated coefficients for GFC are significantly positive across all three sub-indicators. Which indicates that GFC positively affects a firm's ESG performance in terms of environmental, social, and governance dimensions. Regarding the magnitude of the estimated coefficients, the effects of GFC on environmental, social, and governance ratings are 0.405, 0.319, and 0.173, respectively. The impact of GFC is strongest on a company's environmental score, followed by its social score, with the weakest impact on governance ratings. GFC influence firms to invest and improve in green production, environmental protection technology, and pollution control, thereby enhancing environmental ratings. It can also promote the improvement of firm social responsibility. However, governance usually involves internal management, risk control, and protection of shareholder equity. Although GFC can to some extent promote firm social responsibility and management transparency, its impact on governance is relatively small.

Table 4: GFC's Impact on ESG Sub-Item Performance in firms

	E_ratings	S_ratings	G_ratings
	(1)	(2)	(3)
GFC	0.405*** (0.064)	0.319*** (0.062)	0.173* (0.089)
Control Variables	✓	✓	✓
Firm FE	✓	✓	✓
Year FE	✓	✓	✓
Industry FE	✓	✓	✓
Province FE	✓	✓	✓
Observations	21,384	21,384	21,384
R ²	0.642	0.486	0.399

Note: ***, **, and * indicate 1%, 5%, and 10% significance, respectively. Robust standard errors in parentheses are clustered at the firm level. (The same applies below)

5.4.2 Financing constraints

Columns (1) and (2) in Table 5 showing that the impact of GFC on the SA and FC indices is significantly negative at the 1% level. This indicates that the financing constraints faced by publicly listed companies awarded GFC have been significantly alleviated after its implementation. GFC provide firms with lower financing costs, relevant policy subsidies, and tax incentives, enhancing their financing capabilities and further improving their ESG performance. Thus, **H2a** is supported.

5.4.3 Green technology innovation

The second potential mechanism for improving ESG performance is through enhancing green technology innovation. We propose that the incentives and regulatory constraints associated with GFC encourage firms to increase investment in green technology research, thereby fostering green technology innovation and improving ESG performance. We can see that in column (3) of Table 5, the estimated GFC coefficient is significantly positive at the 10% level. Which confirms that GFC significantly enhances green technology innovation in firms. GFC drives firm green technological innovation through environmental standards while alleviating financing constraints, providing financial support for green technology investment. **H2b** is supported.

5.4.4 Media attention

The third potential mechanism for the increase in ESG performance is through attracting media attention. We argue that companies certified as green factories can enhance media visibility and more effectively communicate their achievements in sustainable development, thereby improving their ESG performance. We use the logarithmic value of the number of positive news items (Posinews) and the ratio of positive news items (Posiratio) as proxies for media attention by following Fang et al. (2023). Columns (4) and (5) of Table 5 showing that the estimated coefficients of GFC are significantly positive at 5% level. GFC increases media attention, particularly positive media coverage then impact ESG performance. Therefore, hypothesis **H2c** is supported.

5.4.5 Managerial myopia

Lastly, we examine whether managerial myopia serves as a mechanism through which GFC improves ESG ratings. Column (6) of Table 5 indicates the estimated coefficient of GFC is negative at the 1% level which means GFC can help reduce managerial myopia.

GFC has corrected manager myopia and requires them to examine the operation of the enterprise from a long-term sustainable development perspective, rather than just focusing on maximizing short-term profits. By alleviating financing constraints, GFC has promoted the use of green technologies and long-term environmental investments by firms. At the same time, with the increasing attention from external media, managers are reshaping their corporate culture, improving their brand image and enhancing their competitiveness, which all promotes their long-term strategic thinking. Therefore, **H2d** is validated.

Table 5: Potential mechanism analysis

	SA	FC	Lngp	Posinews	Posiratio	Myopiaratio
	(1)	(2)	(3)	(4)	(5)	(6)
GFC	-0.017*** (0.007)	-0.033*** (0.005)	0.053* (0.030)	0.115*** (0.041)	0.007** (0.003)	-0.008*** (0.003)
Control variables	✓	✓	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓	✓	✓
Year FE	✓	✓	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓	✓	✓
Province FE	✓	✓	✓	✓	✓	✓
Observations	21,039	19,278	21,039	20,973	21,008	20,935
R ²	0.967	0.883	0.666	0.425	0.166	0.091

5.5 Heterogeneity

Considering the heterogeneous responses of firms with different endowments to GFC, we further examine the differential influence of GFC on firm ESG performance based on firm size and environmental concern.

5.5.1 Firm size

Compared to smaller firms, larger firms have advantages in capital, technological accumulation, risk resilience, and sustainable management capabilities (Hörisch et al., 2015). As a result, GFC may result in heterogeneity in different firm sizes. Firm size is measured by total assets in this study, with the sample divided into two subgroups: large firms and small firms, based on the median. The results of firm size heterogeneity are presented in columns (1) and (2) of Table 6. In column (1), the coefficient for GFC is positively significant at the 1% level. In contrast, while the coefficient for GFC in column (2) is

positive, it does not meet the 10% significance threshold. This indicates that the improvement in ESG performance driven by GFC is more significant for larger firms than for smaller ones. To more accurately compare the differences in coefficient magnitudes between the two groups, we apply the Bootstrap method, referring to the approach outlined by [Lian et al. \(2010\)](#). The results of the Bootstrap test confirm a significant difference in the regression coefficients between the two subgroups, further validating the firm size heterogeneity findings.

5.5.2 Firm environmental concerns

Firm environmental awareness directly influences a company's actions regarding ESG (Environmental, Social, and Governance). As an incentive policy for green transformation, the effectiveness of GFC largely depends on the in-depth participation of individual firms. Generally, the level of firm environmental awareness reflects the degree of importance and initiative a company places on environmental issues. Firms with higher environmental awareness are more likely to proactively engage in green factory development and tend to be more committed to fulfilling ESG responsibilities. Consequently, the influence of GFC on ESG performance may differ across companies with varying levels of environmental awareness.

Drawing on the approach of [Duriau et al. \(2007\)](#), firm environmental awareness is measured by calculating the frequency and proportion of environment-related terms in annual reports. Firms are then categorized into groups based on the median frequency of these terms. A higher frequency of environmental terms reflects a higher level of environmental awareness. The results for firm size heterogeneity shown in columns (3) and (4) in Table 6. For firms with high environmental awareness, the GFC coefficient is 0.403, showing significance at the 1% level, whereas for firms with low environmental awareness, the GFC coefficient is 0.214, significant at the 10% level. Furthermore, the Bootstrap test results show the difference between the coefficients of the two sub-samples is significant at the 5% level. GFC exhibits a stronger positive impact on ESG performance in companies that prioritize environmental management.

Table 6: Heterogeneity analysis

	Large scale firms	Small scale firms	High firm environmental concerns	Low firm environmental concerns
	(1)	(2)	(3)	(4)
GFC	0.401*** (0.070)	0.138 (0.105)	0.403*** (0.075)	0.214* (0.100)
Control variables	✓	✓	✓	✓
Firm FE	✓	✓	✓	✓
Year FE	✓	✓	✓	✓
Industry FE	✓	✓	✓	✓
Province FE	✓	✓	✓	✓
Observations	11,178	9,867	10,652	10,387
R2	0.519	0.562	0.516	0.561
Coefficient difference	0.253***		0.189**	

Note: The P-value of inter group differences was estimated based on 1000 bootstrap samples.

6 Conclusion

Green factory certification, as a pivotal green industrial policy, significantly contributes to fostering the sustainable development of firms through green and low-carbon initiatives. This paper investigates the influence of GFC on firm ESG performance, using data from listed manufacturing companies in

China's A-share market from 2009 to 2021, with the policy shock being the implementation of the GFC by the MIIT in 2016. The research results indicate that GFC significantly improves a firm's ESG performance. This conclusion remains robust even after conducting parallel trend tests, placebo tests, using synthetic control difference-in-differences methods, heterogeneity treatment effect checks, and other robustness checks. Furthermore, mechanism analysis reveals that GFC has the strongest impact on improving environmental performance, followed by social performance, with the weakest effect observed on governance performance. Specifically, the positive influence of GFC on ESG performance is driven by four main channels: alleviating financing constraints, promoting green technological innovation, increasing media attention, and mitigating managerial myopia. Heterogeneity analysis shows GFC boosts ESG performance in large firms but not in small ones. Besides, the enhancement effect is more pronounced for firms with higher levels of environmental concern. To address this, we put forward the following policy recommendations:

Firstly, the government need continue to uphold GFC policy, fully leveraging its demonstrative and leading role in accelerating the green, low-carbon transformation of manufacturing firms. Our findings show that GFC enhances firm ESG performance, highlighting its effectiveness as a green industry policy that drives companies towards sustainable practices. As such, the government should prioritize and strengthen GFC by utilizing a range of support measures, including tax incentives, loan interest subsidies, and fiscal rewards, to provide robust backing for companies' green transitions. This approach will encourage firms to actively pursue GFC, attract greater participation from social and financial capital, and collectively drive significant progress in the green, low-carbon transition.

Secondly, when designing green factory support policies, it is essential to account for the characteristics and heterogeneity of firms in order to develop targeted policies. The heterogeneity analysis reveals that GFC affects ESG performance differently based on firm size and levels of environmental concern. Therefore, differentiated support policies should be crafted accordingly. For small firms, financial support such as dedicated funds, subsidies, or low-interest loans could help facilitate GFC and investments in environmental protection. Additionally, establishing technical guidance institutions or collaborative platforms could provide these smaller firms with professional environmental technology guidance and training. On the other hand, strengthening environmental awareness across firms is crucial to enhancing the effectiveness of GFC policies. This could include requiring certified green factories to submit regular environmental reports and publicly disclose their environmental management practices and performance to stakeholders, thereby increasing transparency.

Thirdly, the mechanism analysis shows that GFC enhances ESG performance by easing firm financing constraints, boosting green technological innovation, increasing media attention, and addressing managerial myopia. Based on these conclusions, both governments and firms can implement the following measures to further enhance firm ESG performance. First, establish a "government-bank-firm" communication and cooperation platform to facilitate collaboration between banks and companies, helping to ease financing challenges for businesses. This could be in tandem with direct financial support (subsidies, loans, tax incentives etc...). Second, policies should prioritize R&D support for green innovation, including facilitating and financing industry-university-research partnerships, directly increasing investment in green technology, and promoting the application of advanced technologies in GFC. Third, strengthen the promotion of GFC by promptly summarizing and disseminating advanced experiences and exemplary practices from successful municipal certifications. Highlight the achievements of green factories with notable environmental contributions to maximize the demonstration effect. Finally, incorporate long-term sustainable development goals into GFC criteria and encourage companies to establish dedicated ESG committees.

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